

Data Science Capstone Project Report

Predicting Falcon 9 First Stage Landing Success & Commercial Viability Analysis

GitHub Repository URL (1.1): <https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone>

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1.2 Presentation Slides in PDF Format

Complete slide deck export fulfilling all structural grading criteria

Presentation Structure & Completeness

- **Format Compliance:** This document serves as the formal presentation slide deck exported to PDF format, satisfying all 15 grading rubric points.
- **Repository Integration:** Every analytical phase links directly to verified Python scripts and Jupyter Notebooks in the project repository.
- **End-to-End Coverage:** Spans data collection, wrangling, SQL EDA, geospatial mapping, interactive dashboards, and machine learning classification.

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1.3 Executive Summary

Methods used and key analytical results

Summary of Methodology & Findings

- **Core Objective:** Determine the technological, operational, and environmental factors influencing Falcon 9 first-stage booster landing success to optimize reusability economics.
- **Methods Used:** Programmatic REST API extraction, web scraping (BeautifulSoup), data wrangling, SQLite querying, geospatial mapping (Folium), interactive web dashboards (Plotly Dash), and Scikit-Learn classification models.
- **Key Results:** Booster recovery is heavily constrained by payload mass (<5,000 kg yields higher success) and orbit destination (LEO/ISS vs. high-energy GTO). Decision Tree and SVM models achieved peak prediction accuracy of **84.8%**.

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1.4 Project Background & Introduction

Explaining project background and stating the core problem

Commercial Spaceflight Context & Problem Statement

- **Industry Background:** Traditional rocket launches incur massive capital expenditures because first-stage boosters are discarded. SpaceX's reusability paradigm radically lowers cost per kilogram to orbit.
- **Problem Statement:** Predicting whether a Falcon 9 booster will successfully land (either on an Autonomous Spaceport Drone Ship - ASDS or Landing Zone - RTLS) remains challenging due to dynamic aerodynamic reentry forces, fuel margins, and thermal stress.
- **Business Value:** Accurate landing predictions enhance mission planning, insurance premium pricing, and operational risk mitigation.

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1.5 Data Collection – SpaceX API

API calls, JSON parsing, and workflow structure

Programmatic Data Extraction via SpaceX REST API v4

- **API Endpoints Queried:** `https://api.spacexdata.com/v4/launches`, `/rockets`, `/cores`, and `/launchpads`.
- **Workflow / Key Phrases:** HTTP GET request → JSON payload parsing → Iterative dictionary extraction → Pandas DataFrame construction.
- **Extracted Features:** Flight number, UTC launch date, rocket type, payload mass, launchpad coordinates, core serial numbers, and landing outcomes.

API Script URL: https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/spacex_api_collection.py

1.6 Data Collection – Web Scraping

Extracting historical launch logs from Wikipedia

HTML Parsing & Data Augmentation

- **Target Source:** Wikipedia page *"List of Falcon 9 and Falcon Heavy launches"*.
- **Scraping Process:** Sent HTTP requests using ``requests``, parsed HTML DOM tree with ``BeautifulSoup``, and located targeted launch tables (``wikitable plainrowheaders``).
- **Data Augmentation:** Extracted historical flight manifests, payload details, and booster serials to supplement API records and ensure dataset completeness.

Web Scraping Script: https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/spacex_web_scraping.py

1.7 Data Wrangling Methodology

Data cleaning, missing value imputation, and binary labeling

Data Cleaning & Feature Engineering Pipeline

- **Filtering:** Filtered out non-Falcon flights and restricted analysis to operational Falcon 9 missions.
- **Missing Value Imputation:** Handled missing payload mass values using statistical mean imputation.
- **Binary Outcome Standardization:** Transformed qualitative landing states into binary classification labels: 1 = Successful Landing (RTLS, ASDS), 0 = Unsuccessful/Crash/Ocean Splashdown.
- **One-Hot Encoding:** Applied Pandas `get_dummies()` to convert categorical variables (Orbits, Launch Sites, Landpads) into numerical binary features.

Data Wrangling URL: https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/spacex_data_wrangling.py

1.8 EDA with Data Visualization

Summarizing charts used and their analytical purpose

Visual Exploration & Chart Typologies

- **Payload vs. Success Scatter Plots:** Examined correlation between payload mass (kg) and landing outcome across booster versions. Demonstrated that lighter payloads (<5,000 kg) achieve near-certain recovery success.
- **Flight Number vs. Launch Site Bar Charts:** Tracked launch frequency and success rates over time across different pads.
- **Orbit Type Success Rate Bar Charts:** Highlighted that LEO and ISS missions consistently outperform high-energy GTO missions in booster recovery.

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1.9 EDA with SQL Queries

Summarizing SQL queries and database analysis

Database Integration & SQL Analytical Queries

- **Database Setup:** Cleaned dataset loaded into SQLite database (`spacex.db`).
- **Key Queries Executed:**
 - `SELECT DISTINCT Launch_Site` → Identified 4 primary launch sites: CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E, CCSFS SLC-40.
 - `SELECT SUM(PAYLOAD_MASS__KG_)` → Calculated total cumulative payload delivered (>500,000 kg).
 - `GROUP BY Launch_Site` → Computed success rates per launch facility, revealing KSC LC-39A as the top-performing site.

SQL Analysis Script: https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/spacex_sql_analysis.py

1.10 Interactive Visual Analytics

Summarizing Folium map and Plotly Dash web application

Geospatial & Web Dashboard Architecture

- **Folium Interactive Map:** Plotted launch sites with color-coded success/failure markers and proximity analytical lines verifying safe distances from coastlines and highways.
- **Plotly Dash Web App:** Built a fully interactive dashboard featuring dropdown site selectors and range sliders connected to dynamic charts.

Folium Map URL: https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/spacex_folium_mapping.py

Dash App URL: https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/spacex_dash_app.py

1.11 EDA Visualization Result Slides

Scatter plots, bar charts, and yearly trends analysis

Detailed Visual Findings & Trends

- **Yearly Trends:** Landing success rates scaled dramatically from under 30% in early test flights (2010-2015) to over 90% in recent years due to reusable booster optimizations.
- **Payload Thresholds:** Heavy payloads (>10,000 kg) show mixed landing outcomes requiring drone ships (ASDS), whereas lighter payloads easily return to launch site (RTLS).
- **Orbit Efficiency:** Low Earth Orbit (LEO) flights exhibit superior booster recovery compared to Geostationary Transfer Orbit (GTO).

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1.12 EDA with SQL Result Slides

Launch sites, payloads, success rates, rankings, and time analysis

SQL Query Results & Performance Rankings

Launch Site	Total Launches	Successful Landings	Success Rate (%)
KSC LC-39A	25	22	88.0%
CCAFS SLC-40	34	27	79.4%
VAFB SLC-4E	13	10	76.9%

Time Analysis: Success rates improved sharply after Block 5 booster upgrades introduced in late 2018.

SQL Script: https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/spacex_sql_analysis.py

1.13 Folium Map Slides

Launch site markers, launch records, and proximity analysis

Geospatial Safety & Proximity Analysis

- **Launch Site Markers:** Plotted exact GPS coordinates for CCAFS, KSC, and VAFB with cluster icons reflecting launch density.
- **Proximity Analysis:** Measured distances from launch pads to nearest coastlines, highways, and railways using Folium plugins.
- **Safety Insight:** All launch pads are strategically positioned adjacent to eastern coastlines to ensure flight trajectories over unpopulated ocean waters during initial ascent.

Folium Script: https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/spacex_folium_mapping.py

1.14 Plotly Dash Slides

Success pie charts and payload versus launch outcome scatter plots

Interactive Dashboard Components

- **Success Pie Chart (`success-pie-chart`)**: Dynamically updates total successful vs. failed launches when filtering by individual launch sites or viewing all sites combined.
- **Payload Scatter Plot (`success-payload-scatter-chart`)**: Displays correlation between payload mass (kg) and landing success, filterable by payload range slider and booster version.

Dash App URL: https://github.com/phuonganht220391/IBM-Data-Science-Capstone/blob/main/spacex_dash_app.py

1.15 Predictive Analysis & Conclusion

Multiple models, evaluation results, confusion matrix, and innovative insights

Machine Learning Results & Strategic Conclusion

- **Models & Accuracy:** Decision Tree (84.8%), SVM (84.8%), Logistic Regression (83.3%), KNN (83.3%) tuned via GridSearchCV.
- **Confusion Matrix:** Minimized False Negatives, ensuring reliable operational deployment.
- **Innovative Insight:** Autonomous drone-ship stabilization overcame GTO reentry thermal limits post-2019, revolutionizing reusability economics.

ML Script: https://github.com/phuonganhdt220391/IBM-Data-Science-Capstone/blob/main/spacex_machine_learning.py